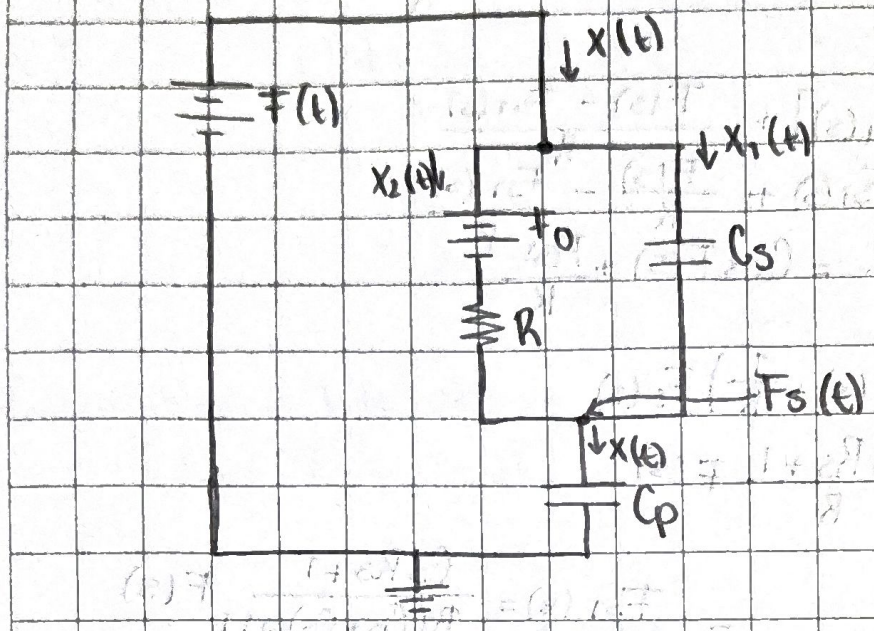
Práctica 3

26 / Oct / 2026

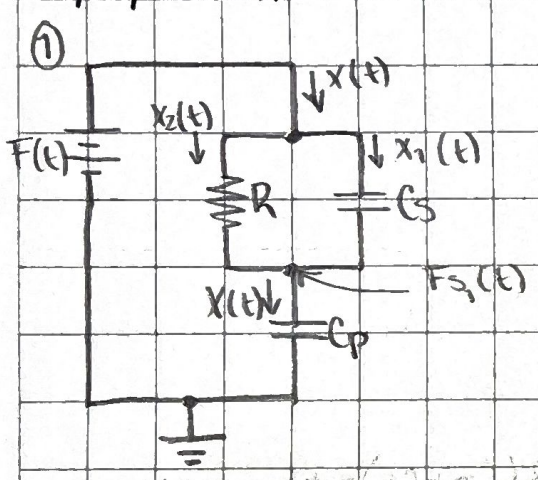
Diagrama

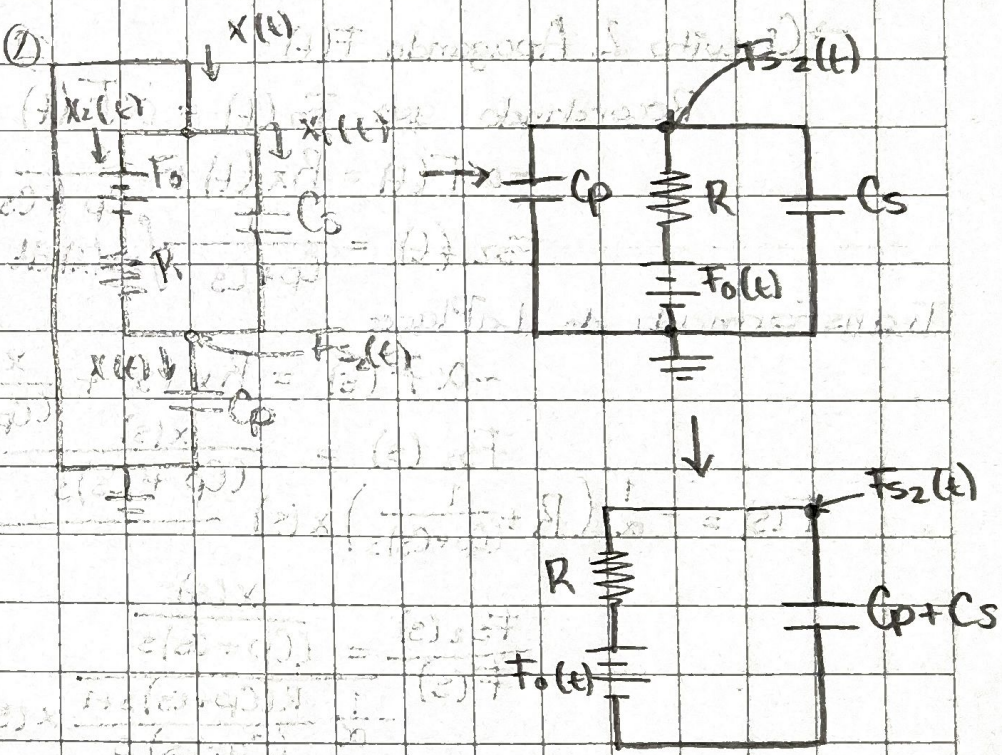


$$x(t) = x_1(t) + x_2(t)$$

$$F_s(t) = F_{s_1}(t) + F_{s_2}(t)$$

Superposición





Análisis matemático

Debido a que el sistema contiene dos fuentes de voltaje, se debe aplicar el principio de superposición

→ Circuito 1. Apagando $F_o(t)$

$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{dF_{s_1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_{s_1}(t)]}{dt}$$

$$x_2(t) = \frac{F(t) - F_{s_1}(t)}{R}$$

$$C_p = \frac{dF_{s_1}(t)}{dt} = C_s \frac{d[F(t) - F_{s_1}(t)]}{dt} + \frac{F(t) - F_{s_1}(t)}{R}$$

Transformada de LaPlace

$$C_p s F_{s_1}(s) = C_s s[F(s) - F_{s_1}(s)] + \frac{F(s) - F_{s_1}(s)}{R}$$

$$C_p s F_{s_1}(s) = C_s s F(s) - C_s s F_{s_1}(s) + \frac{F(s)}{R} - \frac{F_{s_1}(s)}{R}$$

$$C_p s F_{s_1}(s) + C_s s F_{s_1}(s) + \frac{F_{s_1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s_1}(s) = \left(C_s(s) + \frac{1}{R}\right) F(s)$$

$$\frac{R[(C_p + C_s)s + 1]}{R} F_{s_1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s_1}(s) = \frac{\frac{C_s R s + 1}{R}}{\frac{R(C_p + C_s)s + 1}{R}} F(s) = F_{s_1}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

→ Circuito 2. Apagando $F(t)$

Recordando que $F_o(t) = \alpha F(t)$

$$-\alpha F(t) = R x(t) + \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$F_{s_2}(t) = \frac{1}{C_p + C_s} \int x(t)\, dt$$

Transformada de LaPlace

$$-\alpha F(s) = R x(s) + \frac{x(s)}{(C_p + C_s)s}$$

$$F_{s_2}(s) = \frac{x(s)}{(C_p + C_s)s}$$

$$F(s) = -\frac{1}{\alpha}\left(R + \frac{1}{(C_p + C_s)s}\right) x(s) \longrightarrow -\frac{1}{\alpha} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s} x(s)$$

$$\frac{F_{s_2}(s)}{F(s)} = \frac{\frac{x(s)}{(C_p + C_s)s}}{-\frac{1}{\alpha} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s} x(s)} = -\frac{\alpha}{R(C_p + C_s)s + 1} \Rightarrow F_{s_2}(s) = -\frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

Función de transferencia

$$F_s(s) = F_{s_1}(s) + F_{s_2}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$F(s) = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1} F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1}$$

Error en estado estacionario

$$e(s) = \lim_{s\to 0} sF(s)\left(1 - \frac{F_b(s)}{F(s)}\right) = \lim_{s\to 0} s\frac{1}{s}\left(1 - \frac{RC_p s - 1 - \alpha}{(RC_p + RC_s)s + 1}\right)$$

$$= \lim_{s\to 0} s\frac{1}{s}\left(1 - 1 - \alpha\right) = -\alpha$$

Estabilidad en lazo abierto

$$(RC_p + RC_s)s + 1 = \emptyset$$

$$\lambda = -\frac{1}{RC_p + RC_s}$$

El sistema es estable

KYMA